

What is This Project?

This project is a simple Python-based tool that acts as a basic analytics system for transportation data. Instead of tracking expenses, it tracks trips taken by vehicles such as buses, vans, or taxis.

The program reads trip records from a CSV file and helps you understand:

- How many trips were taken
- How much total distance was covered
- How trips are distributed across different vehicles
- How many trips were delayed

The goal was to build a quick, terminal-based way to analyze transport/logistics data, keep it structured, and make the code easy to read and extend.

How It Works (Behind the Scenes)

- **Technology:** Built entirely in Python using only the standard library—no external packages or complex setup required.
- **The Brains:** The program processes each row from a CSV file (like trips.csv) and converts it into a Python dictionary with fields such as trip_id, vehicle_id, distance_km, and delay_min. It then uses simple logic to calculate totals, averages, and counts.
- **Loading Data:** Trip records are stored in a CSV file. The program reads this file, validates each entry (for example, ignoring trips with invalid distance), and keeps everything in memory as a list of trips.
- **The Flow:** You start the program and see a menu in the terminal. From there, you can:
 - Load trip data from a CSV file
 - View a basic summary
 - View per-vehicle trip counts
 - View delay statistics

Each option calls a different function, and the results are displayed neatly on the screen.

- **Code Structure:** The code is split into separate Python modules such as loaders, analytics, reports, and main. This makes the project easy to read, debug, and extend.

Strengths: Why You'd Like It

- **Very Simple to Run:** Works directly in the terminal; you just need Python and a CSV file.
- **Clear Separation of Logic:** Loading, processing, and reporting are handled separately, making the project easy to understand.
- **Realistic Use Case:** Models a real-world scenario of managing trip data for buses, vans, or taxis.
- **Data-Driven Insights:** Quickly see which vehicle is used the most, how many trips were delayed, and

total distance covered.

- Clean and Transparent Code: The logic is readable and easy to explore thanks to modular design.

Conclusion

The TripLog Analyzer Project is a simple, reliable tool for analyzing transportation trip data from CSV files. It demonstrates key Python concepts such as file handling, lists, dictionaries, and modular program design in a practical, easy-to-understand way.